Solving Absolute Value Equations

Beginner Practice Sheet — Study the examples first, then try the problems!

Name: _____ Date: _____ Period: _____

Key Rule — Solving $|E| = c$

Case 1: $c > 0$ $\quad |E| = c \implies E = c$ or $E = -c$ *(two solutions)*

Case 2: $c = 0$ $\quad |E| = 0 \implies E = 0$ *(one solution)*

Case 3: $c < 0$ $\quad |E| = c \implies$ **No solution**

3 Steps: ① Isolate $|\cdot|$ ② Split into two equations ③ Solve each one

Worked Example 1 Solve $|x| = 4$

What it means: $|x| = 4$ says "x is **4 units away from 0**" on the number line. Because distance works in both directions, there are **two answers**.

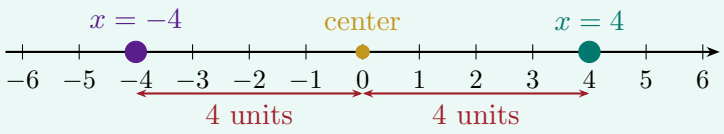
Step ①: The absolute value is already isolated. $|x| = 4$ ✓

Step ②: Split into two equations:

$$x = 4 \quad \text{or} \quad x = -4$$

Step ③: Both are already solved. $\boxed{x = 4 \quad \text{or} \quad x = -4}$

Check on the number line — both dots are exactly 4 units from 0:



Solution set: $\{-4,\ 4\}$

Worked Example 2 Solve $|x-3|+1=6$

What it means: We need to **isolate the absolute value first** (Step ①), then use the distance idea.

Step ①: Subtract 1 from both sides to isolate $|x-3|$:

$$|x-3|+1-1=6-1 \qquad \Longrightarrow \qquad |x-3|=5$$

Step ②: $|x-3|=5$ says "x is **5 units from 3**." Split:

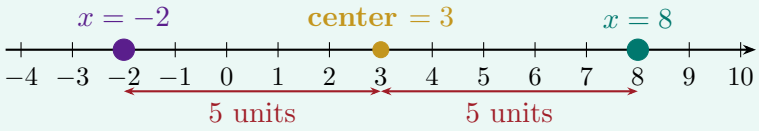
$$x-3=5 \qquad \text{or} \qquad x-3=-5$$

Step ③: Add 3 to each equation:

$$x=8 \qquad \text{or} \qquad x=-2$$

$\boxed{x=8 \quad \text{or} \quad x=-2}$

Check on the number line — both dots are exactly 5 units from the center 3:



Solution set: $\{-2,\ 8\}$

Part A Level 1 — Direct Absolute Value $|x| = c$

Just like Example 1. The answer is $x = c$ or $x = -c$.

1. $|x| = 3$
Step ②: x = _____ or x = _____
Answer: _____

2. $|x| = 8$
Step ②: x = _____ or x = _____
Answer: _____

3. $|x| = 0$
(Hint: $c = 0$, one answer)
Answer: _____

4. $|x| = -4$
(Hint: can distance be negative?)
Answer: _____

5. $|x| = 1$
Step ②: x = _____ or x = _____
Answer: _____

6. $|x| = 10$
Step ②: x = _____ or x = _____
Answer: _____

Part B Level 2 — Shift Inside $|x \pm a| = c$

Split into two cases, solve for x in each.

1. $|x - 2| = 5$

$x - 2 =$ _____ or $x - 2 =$ _____

$x =$ _____ $x =$ _____

Answer: _____

2. $|x + 3| = 4$

$x + 3 =$ _____ or $x + 3 =$ _____

$x =$ _____ $x =$ _____

Answer: _____

3. $|x - 1| = 6$

$x - 1 =$ _____ or $x - 1 =$ _____

$x =$ _____ $x =$ _____

Answer: _____

4. $|x + 5| = 5$

$x + 5 =$ _____ or $x + 5 =$ _____

$x =$ _____ $x =$ _____

Answer: _____

5. $|x - 4| = 0$

(One answer only — why?)

$x - 4 =$ _____ $\Rightarrow$ $x =$ _____

Answer: _____

6. $|x + 1| = 7$

$x + 1 =$ _____ or $x + 1 =$ _____

$x =$ _____ $x =$ _____

Answer: _____

Part C Level 3 — Isolate First! Get $|\cdot|$ alone, then split

Add / subtract / divide to isolate the absolute value first (just like Example 2).

1. $|x| + 3 = 7$

Step ①: $|x| =$ _____

Step ②: $x =$ _____ or $x =$ _____

Answer: _____

2. $|x - 2| + 1 = 6$

Step ①: $|x - 2| =$ _____

$x - 2 =$ _____ or $x - 2 =$ _____

$x =$ _____ $x =$ _____

Answer: _____

3. $2|x| = 10$

Step ①: $|x| =$ _____

Step ②: $x =$ _____ or $x =$ _____

Answer: _____

4. $2|x + 3| = 8$

Step ①: $|x + 3| =$ _____

$x + 3 =$ _____ or $x + 3 =$ _____

$x =$ _____ $x =$ _____

Answer: _____

5. $|x + 2| - 3 = 1$

Step ①: $|x + 2| =$ _____

$x + 2 =$ _____ or $x + 2 =$ _____

$x =$ _____ $x =$ _____

Answer: _____

6. $3|x| = 9$

Step ①: $|x| =$ _____

Step ②: $x =$ _____ or $x =$ _____

Answer: _____

Part D Challenge Show all steps for both cases

1. $|2x - 4| = 6$

Case 1: $2x - 4 = 6$

$2x =$ _____

$x =$ _____

Case 2: $2x - 4 = -6$

$2x =$ _____

$x =$ _____

Answer: _____

2. $|3x + 3| = 9$

Case 1: $3x + 3 = 9$

$3x =$ _____

$x =$ _____

Case 2: $3x + 3 = -9$

$3x =$ _____

$x =$ _____

Answer: _____

3. $|2x + 1| = 5$

Case 1:

_____

_____

$x =$ _____

Case 2:

_____

_____

$x =$ _____

Answer: _____

4. $|4x - 8| = 0$ *Hint: only one answer!*

Answer: _____

Answer Key (Teacher Copy)

Part A

1. $x = 3$ **or** $x = -3$
2. $x = 8$ **or** $x = -8$
3. $x = 0$
4. **No solution**
5. $x = 1$ **or** $x = -1$
6. $x = 10$ **or** $x = -10$

Part B

1. $x = 7$ **or** $x = -3$
2. $x = 1$ **or** $x = -7$
3. $x = 7$ **or** $x = -5$
4. $x = 0$ **or** $x = -10$
5. $x = 4$
6. $x = 6$ **or** $x = -8$

Part C

1. $x = 4$ **or** $x = -4$
2. $x = 7$ **or** $x = -3$
3. $x = 5$ **or** $x = -5$
4. $x = 1$ **or** $x = -7$
5. $x = 2$ **or** $x = -6$
6. $x = 3$ **or** $x = -3$

Part D

1. $x = 5$ **or** $x = -1$
2. $x = 2$ **or** $x = -4$
3. $x = 2$ **or** $x = -3$
4. $x = 2$